

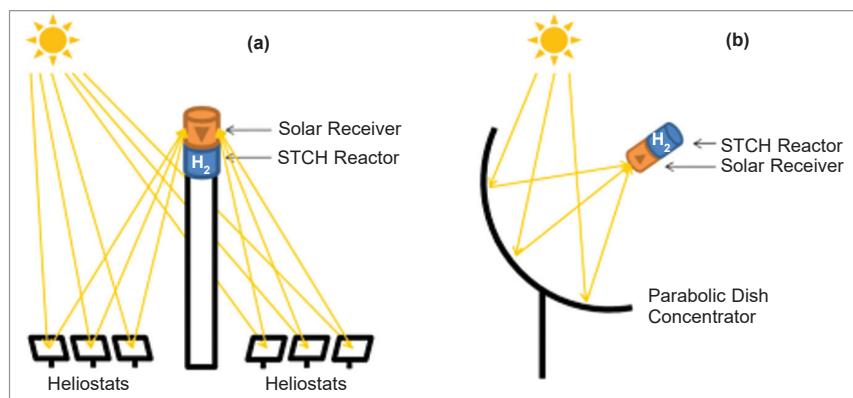


Fig. 5: Thermochemical water splitting process: (a) Central receiver/reactor tower with heliostats, (b) Modular dish-mounted receiver/reactor (Source: <https://www.energy.gov/eere>)

2. Biological processes

Photobiological water splitting process.

The photobiological water splitting processes use microorganisms, such as green microalgae or cyanobacteria, and sunlight to turn water into oxygen and hydrogen ions. The hydrogen ions combine through direct or indirect routes and are released as hydrogen gas. This pathway has low rate of production of hydrogen. The oxygen produced tends to inhibit the hydrogen production and sometimes becomes a safety issue when it mixes with hydrogen.

Photo fermentative process.

In photo fermentative processes, photosynthetic microbes break down the organic matters using sunlight as the driver, releasing hydrogen. At present, the process has a very low hydrogen production rate and low solar-to-hydrogen efficiency, making it commercially unviable for hydrogen production. However, the pathway has a long-term potential, and efforts are on to increase the rate of hydrogen production and make the microbes better at collecting and using energy to improve the efficiency.

3. Microbial biomass conversion processes

The microbial biomass conversion processes utilise the ability of microorganisms to consume and digest biomass and release hydrogen. In fermentation-

based systems, microorganisms, such as bacteria, break down the organic matter to produce hydrogen. The organic matter, which can be used are refined sugars, raw biomass sources such as corn stover, and even wastewater. The microbes break down the complex molecules of the organic matter through many different pathways, and the byproducts are combined by the enzymes to produce hydrogen. In direct hydrogen fermentation systems, the microbes produce the hydrogen themselves. Researchers are trying to make the fermentation systems produce more hydrogen from the same amount of organic matter. As no light is required, these methods are also known as 'dark fermentation' methods.

HOW GREEN HYDROGEN IS TRANSPORTED

Hydrogen is the smallest and the

lightest element (mass of one gallon of gasoline is 2.75kg whereas mass of one gallon of hydrogen is only 0.00075kg, at 1 atm pressure and 0°C). As a fuel, it needs to be delivered from where it is produced to the point of end use, such as an industrial facility, power generator, or fuelling station. Hydrogen in large quantity is transported either in the form of pressurised and compressed gas or as liquefied hydrogen. Usually, pipelines are used to transport hydrogen. Surface transportation of hydrogen involves cryogenic liquid tanker trucks and gaseous tube trailers. Pipelines are deployed in the regions where the demand is substantial and is expected to remain stable for a long time. In the regions where the demand is at a smaller scale or is emerging, the hydrogen is transported using liquid tankers and tube trailers.

The method for transportation is selected based on the production technology used, the scale, the end use, the operating region and the project requirements. For distances below 1500km, transporting hydrogen as a gas via pipeline is the cheapest option. For distances above 1500km, shipping hydrogen as ammonia or via liquid organic hydrogen carriers (LOHC) is more cost-effective. LOHC are preferred where waste heat resources already exist. For distances above 3000km, ships are the most economical mode.

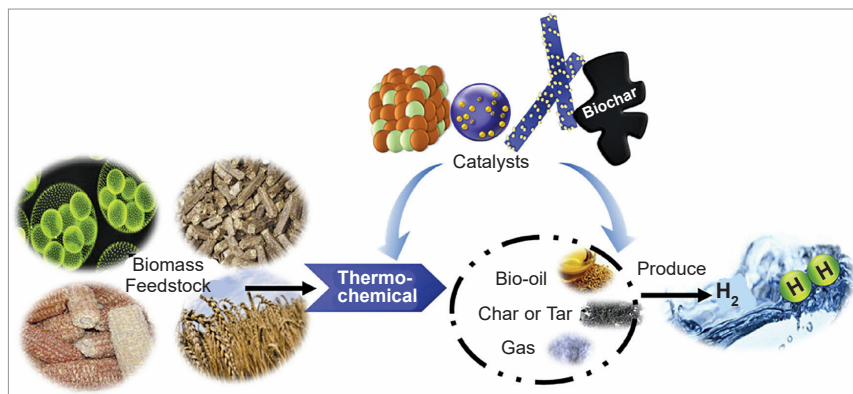


Fig. 6: Hydrogen production from biomass sources (Source: <https://ars.els-cdn.com>)